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Isaac Newton

REASON, OF COURSE, is weak, when measured against its never-ending task. Weak, indeed, compared with the follies and passions of mankind, which, we must admit, almost entirely control our human destinies, in great things and small. Yet the works of the understanding outlast the noisy bustling generations and spread light and warmth across the centuries. Consoled by this thought let us turn, in these unquiet days, to the memory of Newton, who three hundred years ago was given to mankind.

To think of him is to think of his work. For such a man can be understood only by thinking of him as a scene on which the struggle for eternal truth took place. Long before Newton there had been virile minds who conceived that it ought to be possible, by purely logical deduction from simple physical hypotheses, to make cogent explanations of phenomena perceptible to the senses. But Newton was the first to succeed in finding a clearly formulated basis from which he could deduce a wide field of phenomena by means of mathematical thinking, logically, quantitatively and in harmony with experience. Indeed, he might well hope that the fundamental basis of his mechanics would come in time to furnish the key to the understanding of all phenomena. So thought his pupils—with more assurance than he himself—and so his successors, up till the end of the eighteenth century. How did this miracle come to birth in his brain? Forgive me, reader, the illogical question. For if by reason we could deal with the problem of the "how," then there could be no question of a miracle in the proper sense of the word. It is the goal of every activity of the intellect to convert a "miracle" into something which it has grasped. If in this case the miracle permits itself to be converted, our admiration for the mind of Newton becomes only the greater thereby.

Galileo, by ingenious interpretation of the simplest facts of experience, had established the proposition: a body upon which no external force is at work permanently maintains its original velocity (and direction); if it alters its velocity (or the direction of its movement) the change must be referred to an external cause.

To utilize this knowledge quantitatively the conceptions velocity and rate of change of velocity—that is, acceleration in the case of any given motion of a body conceived as dimensionless (material point)—must first be interpreted with mathematical exactness. The task led Newton to invent the basis of differential and integral calculus.

This in itself was a creative achievement of the first order. But for Newton, as a physicist, it was simply the invention of a new kind of conceptual language which he needed in order to formulate the general laws of motion. For a given body he had now to put forward the hypothesis that his precisely formulated acceleration both in magnitude and direction was proportional to the force directed upon it. The coefficient of proportionality which characterizes the body with reference to its power of acceleration completely describes the (dimensionless) body with reference to its mechanical quality; thus was discovered the fundamental conception of mass.

All the foregoing might be described—though in the extremely modest manner of speaking—as an exact formulation of something the essence of which had already been recognized by Galileo. But it by no means succeeded in solving the main problem. In other words, the law of motion yields the movement of a body, only when the direction and magnitude of the force exerted upon it are known for all times. Thus the problem reduced itself to another problem: how to find out the operative forces. To a mind any less bold than Newton's it must have seemed hopeless, considering the immeasurable multifarity of the effects which the bodies of a universe seem to produce upon each other. Moreover, the bodies whose motions we perceive are by no means dimensionless points—that is to say, perceptible as material points. How was Newton to deal with such chaos?

If we push a cart moving without friction on a horizontal plane it follows that the force we exert upon it is given directly. That is the ideal case from which the law of motion is derived. That we are not here dealing with a dimensionless point appears unessential.

How does it stand then with a falling body in space? A freely falling body behaves almost as simply as the dimensionless point, if one regards its movement as a whole. It is accelerated downwards. The